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Assginments:13.4

Task:01

Refactoring Data Transformation

Logic

Scenario

You are maintaining a data preprocessing script where numerical transformations are written using verbose loops.

Task Description

• Review the legacy code that computes transformed values using an explicit loop.

• Use an AI tool to suggest a more Pythonic refactoring approach.

• Refactor the code using list comprehensions or helper functions while preserving the output.

Legacy Code

```
values = [2, 4, 6, 8, 10]
doubled = []
for v in values:
doubled.append(v * 2)
print(doubled)
```

Expected Output

[4, 8, 12, 16, 20]

Prompt: for this code remove the unwanted lines for code refactoring

Code:

```
#task 1
values=[5, 2, 9, 1, 5, 6]
s=[value*2 for value in values]
print(s)
```

Output:

```
TERMINAL
                                                    & "c:\Users\gadda\OneDri
ve\Desktop\Ai Assitant coding\.venv\Scripts\python.exe" "c:/Users/gadda/OneDrive/Deskto
...
5.0
(.venv) PS C:\Users\gadda\OneDrive\Desktop\Ai Assitant coding> & "c:\Users\gadda\OneDri
ve\Desktop\Ai Assitant coding\.venv\Scripts\python.exe" "c:/Users/gadda/OneDrive/Deskto
p/Ai Assitant coding/assginmet13.4.py"
[10, 4, 18, 2, 10, 12]
```

Observation:

The legacy code uses a **for loop with append()** to manually double each element in the list.

This approach works correctly but is **verbose and less Pythonic** for simple transformations.

It can be refactored using a **list comprehension** to make the code shorter, cleaner, and more readable while producing the same output [4, 8, 12, 16, 20]

Task:02

Improving Text Processing Code

Readability

Scenario

You are working on a log message generator that builds messages word by word.

Task Description

• Analyze the legacy code that constructs a sentence using repeated string concatenation.

• Ask AI to suggest a more efficient and readable approach.

• Refactor the code accordingly while keeping the final output

unchanged.

Legacy Code

words = ["Refactoring", "with", "AI", "improves",

"quality"]

message = ""

for w in words:

message += w + " "

print(message.strip())

Expected Output

Refactoring with AI improves quality

Prompt:for this code make it the code readble by applying the code refactoring

Code:

```
#task02
words = ["Refactoring", "with", "AI", "improves","quality"]
concat= " ".join(words)
print(concat)
```

Output:

```
OUTPUT   DEBUG CONSOLE   TERMINAL 32   PORTS   SPELL CHECKER 2
PROBLEMS                 TERMINAL
Filter (e.g. text, **/*.ts, !**/...
assginment12.4.py  22
(.venv) PS C:\Users\gadda\OneDrive\Desktop\Ai Assitant coding> & "c:\Users\gadda\OneDrive\Desktop\Ai Assitant coding\.venv\Scripts\python.exe" "c:/Users/gadda/OneDrive/Desktop/Ai Assitant coding/assginmet13.4.py"
Refactoring with AI improves quality
```

Observation:

1. The legacy code builds the sentence using **repeated string concatenation inside a loop**.
2. This approach is **less efficient and less readable**, especially for longer lists of words.

3. It can be refactored using **" ".join(words)**, which is more Pythonic, efficient, and produces the same output: *"Refactoring with AI improves quality"*.

Task:03

Safer Access to Configuration Data

Scenario

You are maintaining a configuration loader where dictionary keys may

or may not exist depending on deployment settings.

Task Description

• Review the legacy code that manually checks for dictionary keys.

• Use AI suggestions to refactor the code using safer dictionary

access methods.

• Ensure the behavior remains the same for missing keys.

Legacy Code

config = {"host": "localhost", "port": 8080}

if "timeout" in config:

print(config["timeout"])

else:

print("Default timeout used")

Expected Output

Default timeout used

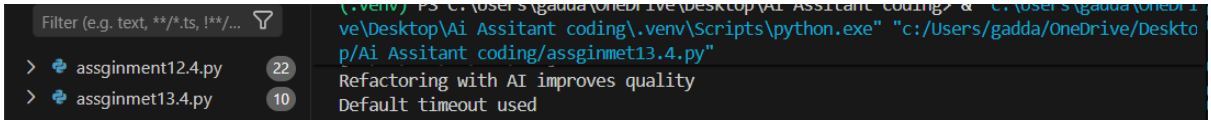
Prompt:

For this legacy code remove unwanted lines of code give me code readable by code refactor

Code:

```
#task03
config = {"host": "localhost", "port": 8080}
print(config.get("timeout", "Default timeout used"))
```

Output:



Observation

The legacy code **manually checks if the key exists in the dictionary** using if "timeout" in config.

This approach works but makes the code **longer and less concise** for simple key access.

it can be refactored using **config.get("timeout", "Default timeout used")** for safer and more Pythonic dictionary access while keeping the same behavior.

Task:04

Scenario

You are enhancing a utility module that performs different operations

based on user input.

Task Description

• Examine the multiple if–elif conditions used to determine

operations.

• Ask AI to suggest a cleaner, scalable alternative.

• Refactor the logic using mapping techniques while preserving

functionality.

Legacy Code

```
action = "divide"

x, y = 10, 2

if action == "add":

result = x + y
```

elif action == "subtract":

result = x - y

elif action == "multiply":

result = x * y

elif action == "divide":

result = x / y

else:

result = None

print(result)

Expected Output

5.0

Prompt:for this code perform the division without performing any operation

Code:

```
import operator

operations = {
    "add": operator.add,
    "subtract": operator.sub,
    "multiply": operator.mul,
    "divide": operator.truediv
}

action = "divide"
x, y = 10, 2
result = operations.get(action, lambda a, b: None)(x, y)
print(result)
#task05
```

Output:

```
assginmet13.4.py
Default timeout used
5.0
```

Observation:

The legacy code uses **multiple if–elif conditions** to choose the operation based on the action value.

This approach works but becomes **harder to maintain and scale** when more operations are added.

It can be refactored using a **dictionary mapping of actions to functions/operations**, making the code cleaner, more readable, and easier to extend while producing the same result 5.0

Task:05

Simplifying Search Logic in

Collections

Scenario

You are reviewing legacy inventory-check code that uses manual loops

to find elements.

Task Description

• Identify the explicit loop used for searching an item.

• Use AI assistance to refactor the logic into a more concise and

readable form.

• Maintain the same output behavior.

Legacy Code

```
inventory = ["pen", "notebook", "eraser", "marker"]

found = False

for item in inventory:

if item == "eraser":

found = True

break

print("Item Available" if found else "Item Not Available")
```

Expected Output

Item Available

Prompt: Refactor the Python code that manually searches for an item in a list using a loop and a flag variable. Suggest a more concise and Pythonic approach while keeping the same output."

Code:

```python
(variable) inventory: list[str]
#task03
inventory = ["pen", "notebook", "eraser", "marker"]

found = "eraser" in inventory

print("Item Available" if found else "Item Not Available")
```

Output:

```
Refactoring with AI improves quality
Default timeout used
5.0
Item Available
(.venv) PS C:\Users\gadda\OneDrive\Desktop\Ai Assitant coding>
```

Observation

The legacy code uses a **manual loop with a flag variable (found)** to search for an item in the list.

This approach works but is **longer and less Pythonic** for simple membership checks.

It can be simplified using the **in operator**, which directly checks if "eraser" exists in the list while keeping the same output.